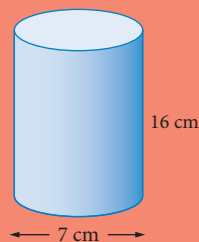


# 2

## MIXED REVISION CHAPTERS 4 • 5 • 6

### Multiple choice

- 1 Which of the following people earns the most?
  - A Peter: \$23 per hour for a 37 hour week
  - B Jessica: commission of 1.5% of \$65 000
  - C Maddy: salary of \$45 000
  - D Mitchell: retainer of \$450 per week plus a commission of 3.5% of \$12 000
  - E Katie: \$4150 a month
- 2  $152 \text{ m}^2$  is equivalent to:
  - A  $1.52 \text{ cm}^2$
  - B  $15\,200 \text{ cm}^2$
  - C  $1.52 \text{ ha}$
  - D  $1\,520\,000 \text{ cm}^2$
  - E  $15\,200\,000 \text{ cm}^2$
- 3 Calculate how much Brad will have in 15 months if he invests \$5000 at 8.3% p.a. compounded quarterly.
  - A \$540.73
  - B \$5540.73
  - C \$12 144.84
  - D \$16 534.72
  - E \$17 144.84
- 4 In December Shannon is paid a bonus of 15% of one month's pay as a Christmas bonus. If she earns a salary of \$72 000, calculate her total pay for the month of December.
  - A \$900
  - B \$6000
  - C \$6900
  - D \$10 800
  - E \$11 700
- 5 Calculate the volume of the following cylinder?
  - A  $352 \text{ cm}^3$
  - B  $616 \text{ cm}^3$
  - C  $704 \text{ cm}^3$
  - D  $1232 \text{ cm}^3$
  - E  $2463 \text{ cm}^3$



- 6 Xavier invests \$40 000 at 7.5% p.a. simple interest. How long will it be before he has \$44 500?  
Answer correct to the nearest month.

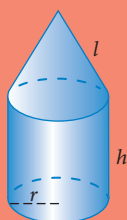
A 2 months  
B 11 months  
C 15 months  
D 18 months  
E 178 months

- 7 Raj works in a convenience store. He is paid normal time if he works Monday to Friday: from 7 a.m. to 5 p.m.; time-and-a-half outside these times and Saturdays; and double time on Sundays. Calculate his pay if he works the following hours and earns \$17.50 per hour.

| Day   | Start time | Finish time |
|-------|------------|-------------|
| Wed   | 5 a.m.     | 1 p.m.      |
| Thurs | 8 a.m.     | 5 p.m.      |
| Fri   | 10 a.m.    | 7 p.m.      |
| Sat   | 10 a.m.    | 7 p.m.      |
| Sun   | 10 a.m.    | 10 p.m.     |

A \$822.50  
B \$831.25  
C \$1111.25  
D \$1146.25  
E \$1233.75

- 8 Which of the following formulas would correctly calculate the surface area of the solid shown?



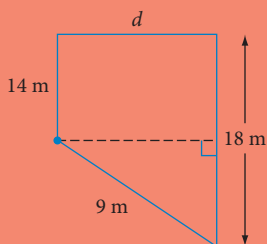
A  $2\pi r^2 + 2\pi rh + 2\pi rl$   
B  $\pi r^2 + 2\pi rh + \pi rl$   
C  $\pi r^2 + 2\pi r^2 h + \pi rl$   
D  $\pi r^2 + 2\pi h + \pi rl$   
E  $\pi r^2 + 2\pi rh + \frac{1}{3}\pi rl$

- 9 Write  $8\frac{1}{4}\%$  p.a. as a decimal per six-months.

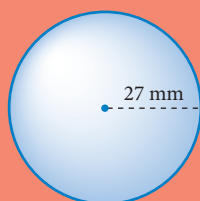
A  $4\frac{1}{8}\%$   
B 0.0418  
C 0.165  
D  $16\frac{1}{2}\%$   
E 0.04125

## Short answer questions

- 1 A company has a total of 1.2 million shares. Its market price is \$3.65 per share and the annual earnings were \$115 000.
  - a Calculate the annual earnings per share, correct to four decimal places.
  - b Calculate the price-to-earnings ratio, correct to nearest whole number.
- 2 Calculate  $d$ , correct to one decimal place.



- 3 How much would you need to put in an investment account to end up with \$15 000 after  $2\frac{1}{2}$  years at 5.6% p.a. compounded monthly? Answer correct to the nearest dollar.
- 4 Steven works a 37 hour week and earns \$34.50 an hour. He is doing up a weekly budget and includes: tax \$320 per week, rent \$350 per week, power \$520 a quarter, groceries \$115 a week, health \$150 a month, fares \$45 a week, phone \$69 per month, clothes \$2400 a year and entertainment \$200 a fortnight. Calculate how much he could save in a year if he sticks to this budget.
- 5 For the sphere shown:

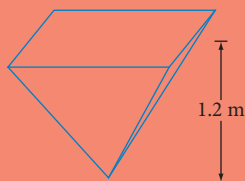


- a Calculate its surface area, correct to one decimal place.
  - b Calculate its volume, correct to one decimal place.
  - c State its capacity to the nearest millilitre.
- 6 Chloe invests \$3200 at 6.24% p.a. simple interest. How much will she have after 8 months?

## Application questions

- 1 A building site manager earns a salary of \$83 561 and a weekly accommodation allowance of \$470.
  - a How much is he paid per fortnight?
  - b He is taking his 4 weeks holiday.
    - i Calculate his annual leave loading if it is 17.5% of 4 weeks of normal pay (with no allowances).
    - ii Calculate his total holiday pay.

- 2 Sandy bought 3500 shares in Fosters at \$6.10 with a dividend yield of 7%. Her stockbroker charges 2.5% brokerage for buying shares and 1% brokerage for selling shares.
- Calculate the total cost to Sandy of buying the shares.
  - Calculate the dividend paid to Sandy.
  - If Sandy sells her shares at \$8.54 per share, one year after buying them, calculate her total profit after costs.
- 3 This grain hopper is in the shape of a square pyramid. It has a volume of  $3.5 \text{ m}^3$  and a depth of 1.2 m.



- What is the area of its square top?
  - What is the side length of the square top (correct to three decimal places)?
  - What is the slant height of the hopper (correct to three decimal places)?
  - How many square metres of stainless steel would have been used to create the surface of the hopper, assuming that there is no lid on the hopper? Round your answer to one decimal place.
- 4 Tony's fish tank is 1.2 m long, 80 cm wide and 90 cm high. Its four walls and base are made from glass. The top is made of Perspex.
- Calculate the total amount of glass surface in this fish tank to the nearest square centimetre.
  - What is the volume of the tank in cubic metres?
  - What is the capacity of the tank in litres?
  - Tony has chosen fish which are approximately 3 cm in length. He has been advised that each of these fish will require 6 litres of water. How many fish will he put in his fish tank if he follows the recommendation?
- 5 Olivia borrowed \$15 000 at 6.5% p.a. flat rate interest.
- If the loan was for 3 years, how much interest did she have to pay?
  - What was the total amount she had to repay?
  - If she paid the loan back in monthly instalments, what was the amount of each instalment? Answer correct to the nearest cent.
- 6 Samuel is at university when he inherits \$75 000. He wants to look into which account at his bank is best. He has three options:
- Option A: 7.65% p.a. simple interest  
 Option B: 7.1% p.a. compounded monthly  
 Option C: 6.95% p.a. compounded daily
- He initially wants to invest it for 2 years until he finishes university. Calculate the amount of interest he will earn for each option and hence decide the best option.
  - He then changes his mind and decides that he will invest it for 3 years to give himself time to get sorted. Does this change his decision on which account? Explain your reasoning.